

Nabla v3

A weekly total-return model with a cash dial

October 3, 2026

Summary. A concentrated, slightly high-beta 15-stock book built from five point-in-time factors, rebalanced weekly on new data, with one regime rule that moves to cash in stress. No options. Nothing in this plan has been run on the data yet; every performance statement is a hypothesis to test.

1 Ground rules

The organizers' answers decide the design. Record them in `docs/AUDIT.md` with who answered and when, so a judge does not read the full-data fit as holdout contamination.

Organizer answer	What v3 does
Cash is allowed	A regime cash dial replaces puts. No options.
Scored mostly on total return; other metrics still count	Lean aggressive but not a lottery ticket: beta 1.1–1.2, 15 names.
Transaction costs must be in our model	Every backtest trade pays a modeled cost.
Data stops at the cutoff; test is the next 30 days	Fit on all history; trade live on new weekly data.
Weekly rebalancing allowed	Model scores live each week; no refitting during the test.
We set our own position and sector limits	Limits in Section 3, each justified in <code>AUDIT.md</code> .

Why no options. Weekly cash moves hedge downside at no premium. Puts bleed premium in flat months. Calls lose on average because implied volatility usually exceeds realized volatility.

2 Signals

Five factors, each with an economic reason fixed before any testing. The default model is an equal-weight composite; ridge regression is kept only if it beats the composite out of sample.

Factor	Construction	Why it should work	Data
Momentum 12-1	Return from $t - 12m$ to $t - 1m$ on split- and dividend-adjusted prices	Investors underreact; trends persist for months	<code>stocks_daily</code> , <code>corporate_actions</code>
Guidance velocity	<code>guidance_range_velocity</code> : same-period guidance raises or cuts per day	Management revisions trend and lead price, like analyst estimate revisions	State vector (PIT Benzinga timestamps)
Quality	ROE or gross profit / total assets from filed fundamentals	Profitable firms outperform and hold up in selloffs	<code>ds.asof()</code> (filing-date gated)
Value	Earnings yield: filed TTM EPS / price	Cheap stocks outperform; offsets momentum crashes	<code>ds.asof()</code> + prices
Volatility premium	$-\log(\text{IV}_{30}/\text{RV}_{21})$	Options priced above delivered volatility predict lower returns (Bali and Hovakimyan, 2009)	<code>atm_iv</code> ; realized vol from <code>microstructure</code> averaged over 21 days

Construction

For factor k , stock i and date t , with moments taken across stocks within sector group g :

$$z_{i,t}^{(k)} = \text{clip}\left(\frac{x_{i,t}^{(k)} - \mu_{g,t}^{(k)}}{\sigma_{g,t}^{(k)}}, -3, 3\right), \quad s_{i,t} = \frac{1}{|K_{i,t}|} \sum_{k \in K_{i,t}} z_{i,t}^{(k)},$$

where $K_{i,t}$ is the set of factors available for stock i on date t .

1. Winsorize each raw factor at the 1st and 99th percentiles across stocks on each date.
2. Z-score within about 10 sector groups mapped from SIC division, not the raw `sic_description` (hundreds of categories).
3. A missing factor is treated as neutral; the dataset's null flags are kept for diagnostics.

Dropped from earlier plans. `fundamental_surprise` compares full-year revenue guidance with TTM actuals, so it measures guided growth rather than beats. The raw `variance_risk_premium` uses one day of realized volatility. `composite_valuation_gap` has no filing-date gate and largely duplicates one-year price reversal. Amihud illiquidity moves to the liquidity filter and cost model.

3 Portfolio rules

Hold the top 15 names by composite score after the liquidity filter, weighted by score, inside these limits.

Rule	Value	Reason
Liquidity filter	20-day ADV > \$50M, price > \$5, most illiquid Amihud decile excluded	Every name must be tradable with small cost
Number of names	15	Concentrated enough to beat a diversified field; one miss is survivable
Max per name	10%	One blowup costs at most about 3–5% of the book
Max per sector group	30%	Avoids a single-theme bet
Target beta	1.1–1.2	The equity premium rewards exposure; higher beta gives up too much Sharpe
Earnings in window	Max 6% per reporting name	Earnings gaps of 10–20% are common; no combined cap, since most names report in October
Size vs liquidity	Max 1% of 20-day ADV	Keeps market impact negligible on \$1M
Turnover band	Trade only if a target weight moves more than 2 points	Prevents weekly churn

Beta tilt. Break near-ties in composite score toward higher shrunk beta, $\tilde{\beta}_i = 0.67 \hat{\beta}_i + 0.33$, with $\hat{\beta}_i$ estimated on 252 daily returns against SPX. Do not optimize for beta directly, or the book becomes a pure beta bet.

Crowding check. Compare the top 15 with a naive 12-month momentum top 20. Heavy overlap means the book holds what most other teams hold, which lowers the chance of finishing ahead of them.

4 Weekly loop

Fit once before the deadline, then only score each week. Nothing is refit during the test.

Offline, before the deadline

1. Fit factor weights (equal weight, or ridge if it wins), beta estimates and cash-dial thresholds on all history.
2. Save them to `config/model.json` with the git commit and a data hash.
3. Cache slow inputs: the 12-month momentum base, filed fundamentals and betas.

Live, each rebalance day

1. Set the as-of date to the last trading day in the data, `max(ds.trading_days())`, never `date.today()`.

2. Load only the newest rows and update the cached inputs.
3. Compute factor z-scores and the composite.
4. Check the cash dial.
5. Pick 15 names and apply the limits and turnover band.
6. Return weights from `/portfolio/holdings` within the 30-second limit.

If no new data arrives, step 1 returns the same date and the book is unchanged. That is the safe fallback.

5 Cash dial

One rule, checked weekly. In a simulation against a high-beta field of 25 teams, a defensive book with 30% cash won about 25% of months in which the market fell more than 3%, against under 5% for aggressive books. That is the reason to build it.

State	Trigger	Book
Normal	Either stress condition is false	0–5% cash, beta 1.1–1.2
Stress	SPX below its 200-day average <i>and</i> 21-day market realized volatility or <code>funding_stress</code> above its 80th percentile	25% cash; replace the highest-beta names with the next-best lower-beta names

- **Hysteresis.** Leave Stress only after both conditions have been false for two straight weeks.
- **Fixed thresholds.** Set them before testing and test once. Keep the rule only if it cuts maximum drawdown without lowering total return by more than about 1 point a year.
- **Rubric check.** The public `check.py` still requires weights to sum to 1 ± 0.01 . If the judge has not changed this, Stress holds low-beta names instead of cash.

6 Costs

Every trade in the backtest pays a half-spread plus market impact on the dollar amount traded:

$$c_i = |\Delta w_i| V \left(\frac{1}{2} s_i + k A_i |\Delta w_i| V \right),$$

where V is portfolio value (\$1M at the start), Δw_i the weight change, s_i the quoted spread estimate, A_i the 21-day average Amihud illiquidity (price move per dollar traded), and $k = 1$ to start.

- Spread s_i : 5 bps for the most liquid third of names by ADV, 10 bps for the middle third, 20 bps for the rest.
- Sensitivity: rerun at twice these costs; the strategy must still beat the baselines.
- Live: the first week trades the full book once; later weeks trade only names past the turnover band.

At \$1M in names above \$50M ADV, expect roughly 5–15 bps per full rebalance. Report the measured figure in `AUDIT.md`.

7 Validation

The backtest runs the live loop exactly: weekly rebalancing, the same limits, costs charged, from 2018 to the cutoff.

Setup

- Walk-forward: anything fitted uses only data before each rebalance date; labels lag 21 trading days.
- The last 6 months are held back untouched until the final go/no-go, then everything is refit on all data.
- One return function matches `check.py`; `/backtest` reuses it, and it is unit-tested on random weights.

Metrics, against SPX and an equal-weight liquid universe

- Total return, annualized return, Sharpe ratio, maximum drawdown, turnover, total cost paid.
- Rolling 21-day returns: median, 5th and 95th percentiles, worst window.
- Sub-periods: 2018 Q4, 2020, 2022.

Leak and robustness tests

- Shuffled labels: the composite's edge should fall to about zero.
- Signals delayed one day: performance should fall.
- Each factor alone, and the composite minus each factor, to show every factor earns its place.
- Twice the costs: still beats the baselines.

Survivorship. The universe is today's survivors, which flatters concentrated, high-beta books. State this in `AUDIT.md` and never present backtest returns as expected returns.

8 Build order

Each step must work before the next starts. If time runs out, ship the last step that passed.

When	Ezekiel (quant)	Jesse (finance)	Done when
Sat 6–8 PM	Loader anchored to the last data date; split and dividend adjustment; deploy the starter and pass <code>check.py</code>	Factor formulas; SIC-to-sector map; liquidity filter values	NVDA 2024 and AAPL 2020 splits look clean; public URL scores 100
Sat 8 PM–midnight	Weekly backtest harness with the cost model; composite and portfolio rules	Check each factor’s sign and IC; review sample holdings	Beats the equal-weight universe after costs; shuffle test passes
Sun 12–2 AM	Cash dial test; held-back 6-month check; freeze <code>model.json</code>	Charts: equity curve, drawdown, sector exposure vs SPX	Go/no-go decided on the held-back period
Sun morning	Live weekly scoring in <code>/portfolio/holdings</code> ; redeploy; rerun <code>check.py</code>	AUDIT.md (factor rationale, limits, organizer answers, survivorship); pitch	Endpoint returns this week’s book in under 30 s; repository public

9 Cut list and open questions

Not building: options (puts or calls), reinforcement learning, LightGBM, Black-Litterman or CVaR optimizers, and any earnings-event overlay beyond the per-name cap.

Questions for the organizers

1. How does new data arrive during the test: does the dataset server update, or does the judge call our endpoint with a date?
2. On which day of the week is the book read, and is the 30-day return compounded from the weekly holdings?
3. Does the holdings check accept weights summing below 0.99 now that cash is allowed?
4. Which metrics count besides total return (Sharpe, drawdown), and with what weight?